

Overfitting

Project: Responsible Plant Classifier

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Reproducible Experiments

Without opening the old lesson, explain one key idea from Reproducible Experiments. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Train, Validation, Test

Without opening the old lesson, explain one key idea from Train, Validation, Test. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Absolute Error and MAE

Without opening the old lesson, explain one key idea from Absolute Error and MAE. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Interpret a gap for Overfitting and model capacity. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Overfitting and model capacity. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Changed independent case

Model A training 96%, validation 62%; model B training 88%, validation 79%. Compare gaps and choose a development candidate, assuming the same valid evaluation conditions.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

A has training 98%, validation 58%; B has training 86%, validation 81%. Compute gaps and select a candidate with a qualification.

6. Solve the given case

A scores 100% train, 60% validation; B scores 85%, 80% on the same sets.

Compute gaps and select by validation accuracy.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

Model A has training accuracy 99% and validation accuracy 70%; B has 88% and 84%. Compute each gap and discuss the evidence.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 27 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Check Plant Classifier train-validation gaps and limit complexity before final test release.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

A has training 98%, validation 58%; B has training 86%, validation 81%. Compute gaps and select a candidate with a qualification.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Select from evidence

Choose among the three worked candidates and explain the training trap.

My reasoning and evidence

13. W2 Interpret a gap

Does a training-validation gap alone prove overfitting?

My reasoning and evidence

14. W3 Propose a controlled change

Describe one fair next development comparison.

My reasoning and evidence

15. W4 Apply independently

Models A training 4 validation 5, B training 1 validation 2, C training 0 validation 2. Lowest validation MAE then simpler; B is simpler than C. Choose and justify.

My reasoning and evidence

16. W5 Limit a diagnosis

What extra evidence would you inspect before concluding a model memorized noise?

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Overfitting

See overfitting as a gap requiring investigation, not a label based only on model size.

Toy model Small: training 85%, validation 80%. Large: training 100%, validation 60%. Same split and scoring rule.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Training and validation accuracy

percent



Bar length shows the supplied value; the marked baseline is zero.

Overfitting

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compute gaps

Small gap=5 percentage points; Large gap=40 percentage points.



2. Select development candidate

Small has the higher validation accuracy despite its lower training fit.



3. Investigate large model

Check capacity, training choices and data mismatch without tuning on final cases.

CHANGE ONE THING

Large's validation accuracy rises to 82% after a development revision.

Explain the changed result using the matching lesson.

Does 100% training accuracy prove Large was always best? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

19. Project portfolio milestone

Project: Responsible Plant Classifier

Current milestone: Prepare the demonstration. Gather reproducible instructions, evidence, access checks and remaining limitations.

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

See overfitting as a gap requiring investigation, not a label based only on model size.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Check capacity, training choices and data mismatch without tuning on final cases.

1. Select development candidate
2. Compute gaps
3. Investigate large model

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Investigate large model Check capacity, training choices and data mismatch without tuning on final cases. Read the picture in order. Check capacity, training choices and data mismatch without tuning on final cases. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed-case practice: Large's validation accuracy rises to 82% after a development revision. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. This one example guarantees correct results for every future input.
3. Its gap is 18 percentage points; it now beats Small on this validation set.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Its gap is 18 percentage points; it now beats Small on this validation set. Validation performance improved, but the training-validation gap is not a complete release rule.

Repair a missing building block

Review Reproducible Experiments (ai-4/26). Goal: Record enough detail to reproduce a run. Short practice: What should accompany the mean 70%? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Train, Validation, Test (ai-4/24). Goal: Select a model using a declared validation rule. Short practice: Why does B's zero training MAE not settle the original choice? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Absolute Error and MAE (ai-4/21). Goal: Calculate MAE MSE and RMSE. Short practice: Why is quoting MSE as 4.67 cm misleading? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

